

TAOLAGNARO, Madagascar

WMO: 671970

Lat: 25.038S

Lon: 46.956E

Elev: 9

StdP: 101.22

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.1	16.1	10.9	8.1	20.0	12.0	8.7	19.7	12.3	23.0	11.3	22.9	1.1	320	0.421

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	6.2	32.1	25.7	31.6	25.5	31.0	25.4	26.8	30.3	26.4	29.8	26.0	29.3	7.4	60

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.9	21.3	29.1	25.2	20.4	28.3	25.1	20.2	28.1	84.3	30.7	82.5	30.0	80.8	29.7	29.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
13.3	12.0	10.7	DB	12.9	34.3	0.9	1.1	12.2	35.0	11.6	35.7	11.1	36.3	10.4	37.0
			WB	11.8	28.2	0.9	0.8	11.1	28.8	10.6	29.3	10.1	29.7	9.5	30.3

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	24.2	27.1	27.0	26.5	25.0	23.0	21.6	20.9	21.5	22.5	23.9	25.0	26.6
	DBStd	2.71	1.42	1.31	1.41	1.69	1.54	1.49	1.52	1.47	1.67	1.78	1.78	1.67
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	2	0	0	0	0	0	0	1	1	0	0	0	0
	CDD10.0	5186	531	477	512	451	403	347	337	358	377	431	449	514
	CDD18.3	2146	273	244	254	201	144	97	79	99	127	172	199	256
	CDH23.3	14849	2594	2321	2200	1334	578	272	174	307	542	936	1334	2259
	CDH26.7	3714	782	678	577	269	56	15	5	21	72	196	349	693
(14) Wind	WSAvg	5.1	5.4	5.1	5.0	4.3	3.9	3.8	4.2	5.3	6.2	6.2	6.1	5.6
(15) Precipitation	PrecAvg	1642	236	216	200	162	136	86	120	91	37	84	108	163
	PrecMax	2577	995	647	753	423	344	182	519	236	102	270	498	407
	PrecMin	1054	56	16	18	22	14	0	5	1	2	6	1	33
	PrecStd	400	213	166	161	99	86	54	103	60	29	71	102	109
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	33.1	32.8	32.3	31.2	29.2	28.1	27.2	28.2	29.5	31.0	32.0	32.9
		MCWB	26.4	26.1	25.9	25.2	23.4	22.0	20.8	21.2	22.7	24.1	24.8	25.8
	2%	DB	32.1	31.8	31.2	30.2	28.0	26.9	26.1	27.1	28.2	29.8	30.8	32.0
		MCWB	25.8	25.7	25.6	24.6	23.0	21.5	20.5	21.0	21.8	23.6	24.4	25.6
	5%	DB	31.2	31.0	30.4	29.2	27.1	25.9	25.1	26.1	27.4	28.8	29.9	31.1
		MCWB	25.7	25.6	25.3	24.2	22.5	21.0	20.1	20.6	21.4	22.9	24.0	25.3
	10%	DB	30.2	30.1	29.8	28.2	26.1	25.0	24.2	25.1	26.4	27.8	28.9	30.1
		MCWB	25.3	25.4	25.0	23.8	22.1	20.5	19.8	20.2	21.0	22.5	23.6	25.1
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	27.6	27.5	27.0	26.3	24.9	23.1	22.2	22.8	23.7	25.3	26.2	27.0
		MCDB	31.3	30.6	30.7	29.9	27.5	26.4	25.5	26.4	28.1	29.7	30.3	31.2
	2%	WB	26.7	26.7	26.4	25.6	24.0	22.4	21.5	22.0	22.7	24.2	25.3	26.3
		MCDB	30.2	30.0	29.7	28.8	26.7	25.4	24.6	25.7	26.7	28.1	29.3	30.2
	5%	WB	26.3	26.2	25.9	25.0	23.3	21.7	20.8	21.3	22.2	23.6	24.6	25.9
		MCDB	29.6	29.4	29.0	28.0	26.0	24.7	23.9	24.7	26.0	27.2	28.4	29.6
	10%	WB	25.9	25.8	25.5	24.4	22.6	21.1	20.4	20.7	21.6	23.0	24.0	25.4
		MCDB	29.0	28.8	28.5	27.3	25.3	23.9	23.3	24.0	25.2	26.4	27.5	29.0
(35) Mean Daily Temperature Range	5% DB	MDBR	6.2	6.1	6.0	6.3	6.6	6.9	6.9	7.3	7.3	7.0	6.7	6.5
		MCDBR	7.3	7.2	7.0	7.2	7.6	8.4	8.3	8.9	8.9	8.4	8.3	7.7
		MCWBR	3.3	3.3	3.3	3.6	4.1	4.7	4.6	5.0	4.6	4.2	4.0	3.5
	5% WB	MCDBR	6.7	6.4	6.4	6.8	6.9	7.6	7.7	8.3	8.4	7.8	7.7	7.2
		MCWBR	3.4	3.3	3.3	3.7	4.2	4.6	4.8	5.0	4.8	4.2	4.1	3.7
(40) Clear-Sky Solar Irradiance	taub		0.421	0.419	0.409	0.386	0.361	0.344	0.344	0.364	0.409	0.459	0.436	0.425
	taud		2.465	2.467	2.496	2.541	2.594	2.605	2.593	2.509	2.358	2.247	2.358	2.431
	Ebn at Noon		925	911	892	869	851	847	860	877	877	865	905	922
	Edh at Noon		120	117	108	95	83	78	82	97	122	144	132	124
(44) All-Sky Solar Radiation	RadAvg		6.81	6.40	5.72	4.74	3.94	3.53	3.70	4.61	5.72	6.31	6.78	7.09
	RadStd		0.49	0.43	0.38	0.22	0.13	0.08	0.11	0.22	0.22	0.37	0.34	0.34

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	+0.20	N/A	+0.92	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page